After connecting and , the currents in and are

Thus connecting , leaves the currents in , unchanged.
Now only , act; , are shorted. Find each branch
current. Opening , gives the circuit shown. As
above, with , alone, currents in , are

The currents in and are

Across and , . On reconnecting , , currents in and
are

Thus connecting , leaves all branch currents unchanged.
By superposition, the currents in and are

As in Method .
Scoring: use the criteria for Method .

VIII.
Solution:
Below, rays reflected by the spherical mirror meet at the convex lens focus.